

MOGFormer: A Patient-Specific Multi-Omics Graph Transformer with Hierarchical Pathway Encodings for Interpretable Cancer Subtyping

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Abstract

Accurate stratification of cancer subtypes is essential for personalized therapy, yet deep models struggle in the $N \ll p$ regime characteristic of multi-omics cohorts, where well-tuned classical early-fusion models remain formidable baselines. Multi-modal fusion methods also suffer from information routing biases, lazily relying on high-density transcriptomics while ignoring epigenetic and copy-number signals, and standard graph neural networks fail to respect established biological topographies. We present MOGFormer, a patient-specific multi-omics graph transformer that builds unified gene-level tokens from raw multimodal continuous signals through non-linear feature tokenizers and localized intra-gene cross-attention, and structures its global attention with a soft inductive bias derived from Protein-Protein Interaction (PPI) shortest-path distances, read out through specialized Latent Pathway Tokens. We evaluate MOGFormer on 949 TCGA-BRCA patients (RNA-seq, copy-number variation, methylation) for PAM50 subtyping under a deliberately rigorous, leakage-free protocol: repeated stratified patient-level cross-validation with all feature selection, normalization, and graph construction re-fit inside each fold, and paired comparison against a full classical baseline suite. Beyond predictive performance, MOGFormer offers native, multi-tiered interpretability, mapping global patient states back to single-gene multimodal driver mechanisms without post-hoc explainers.

Introduction

In the era of precision oncology, leveraging machine learning for *in silico* biological discovery requires architectures that simultaneously navigate vast feature spaces and yield mechanistic insight. Traditional early-fusion models often match or outperform deep networks on multi-omics cohorts because of the acute $N \ll p$ regime, where the absence of structural constraints lets flexible models overfit [6, 8]. To embed biological reality into deep models, Graph Neural Networks (GNNs) have been used with Gene Regulatory Networks or PPI networks as structural priors. However, message-passing GNNs suffer from two architectural flaws: the over-squashing of information during localized aggregation, and an inability to discover *de novo* or context-specific interactions beyond the static edges of the input graph.

To transcend these limitations, we extend the transformer architecture to biological graphs through MOGFormer. Following the Graphormer paradigm [3], global self-attention parametrized by a soft structural bias prevents information bottlenecks while letting the network learn long-range genetic dependencies absent from static databases.

Crucially, cellular state is not driven by transcriptomics alone; it is governed by a multi-tiered epigenetic and structural landscape. MOGFormer captures three primary modalities:

- **RNA-sequencing (mRNA):** the high-density downstream transcriptional output and functional state of the cell.
- **Copy Number Variations (CNV):** discrete structural alterations (focal amplifications, deletions) that set baseline expression boundaries.
- **Methylation β -values:** localized CpG-island modifications that dynamically silence or activate transcription.

Intra-Gene Interactions: Epigenetic & transcriptomic

Cellular identity is not dictated by a flat vector of independent genomic features, but by a hierarchical regulatory cascade occurring within each individual gene. While transcriptomics provides a high-density snapshot of the cellular functional state, it is bounded by structural alterations (CNV) and dynamically regulated by epigenetic modifications (methylation). Conventional early-fusion models simply concatenate these signals, implicitly treating a gene’s promoter methylation as independent of its own transcriptional output. However, biological reality dictates localized cross-talk—for instance, hypermethylation directly silences mRNA expression. To faithfully capture this regulatory grammar, an architecture must explicitly resolve these intra-gene dependencies before evaluating the global cellular state. This localized biological constraint directly motivates our intra-gene cross-attention mechanism, which learns a unified, mechanistically coherent token for each gene.

Inter-Gene Interactions: Navigating Pathway Topographies

Beyond localized regulation, oncogenic phenotypes are driven by systemic dysregulations across complex biochemical networks. Genes and their downstream protein products do not operate in a vacuum; they interact through structured signaling cascades and physical complexes. While static Protein-Protein Interaction (PPI) networks map these potential topological connections, standard message-passing GNNs applied to these structures suffer from information over-squashing and fail to capture context-specific, long-range dependencies that emerge *de novo* in tumor states. Conversely, naive multi-omics transformers ignore established biological topology entirely, leading to spurious correlations in the $N \ll p$ regime. This necessitates a hybrid approach: a structurally biased global attention mechanism where established PPI shortest-path distances act as a soft inductive prior. This allows the network to route information along known canonical

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Figure 1 placeholder — PAM50 class distribution (the imbalance), 166/77/480/194/35.

Figure 1: PAM50 subtype distribution across the 949-patient TCGA-BRCA cohort.

pathways while retaining the flexibility of global self-attention to discover novel inter-gene dependencies.

Rather than treating each of the three modalities as disjoint vectors, our architecture explicitly models their intra-gene cross-talk (e.g. methylation-driven silencing of expression) prior to global state computation.

We validate this framework on the The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort [2]. Breast cancer is highly heterogeneous, conventionally classified via the PAM50 panel [1] into Basal-like, HER2-enriched, Luminal A, Luminal B, and Normal-like subtypes, which carry distinct prognosis and therapeutic implications. While robust classification matters, the core contribution of MOGFormer lies in its native, multi-tiered interpretability: bypassing post-hoc explainers (LRP, SHAP), the routing mechanics expose which global pathways drove a classification, which driver genes anchored those pathways, and which omics modality dictated an anomalous gene state.

1. Dataset and Preprocessing

1.1. Dataset

We use TCGA-BRCA, retaining female patients with matched RNA-seq, copy-number, and methylation β -value profiles for the genes of interest, yielding 949 patients. The PAM50 label distribution is markedly imbalanced: 166 Basal, 77 HER2, 480 Luminal A, 194 Luminal B, and 35 Normal-like (Fig. 1). This imbalance, in particular the 35 Normal-like cases, governs both the evaluation design (Sec. 4) and the dominant error modes (Sec. 5).

1.2. Leakage-Free, Per-Fold Preprocessing

Preventing data leakage is critical and is enforced *within* every cross-validation fold rather than once globally. Let the raw inputs be $\mathbf{X}_{mRNA}$, $\mathbf{X}_{CNV}$, $\mathbf{X}_{methy}$ and the clinical vector $\mathbf{Y}$. Only patients present across all three modalities and the clinical table are retained, deduplicated at the patient level. Every statistical operation below is fit on the training fold alone and merely applied to held-out patients.

Highly Variable Gene (HVG) Selection. To extract informative genes while remaining robust to outliers, we rank by the Median Absolute Deviation (MAD),

$$\text{MAD}(\mathbf{x}) = \text{median}(|\mathbf{x} - \text{median}(\mathbf{x})|), \quad (1)$$

computed per modality, and combine modalities through a consensus rank

$$\text{Rank}_{\text{cons}} = \text{Rank}_{mRNA} + \text{Rank}_{CNV} + \text{Rank}_{methy}. \quad (2)$$

The top N genes by consensus rank are kept. A curated set of PAM50/Prosigna and known BRCA driver genes is *force-included* after ranking, guaranteeing that canonical markers (e.g. *ESR1*, *ERBB2*, *MKI67*) are present in every fold regardless of their MAD rank. Unless otherwise stated we use $N = 250$ MAD genes augmented by the curated set.

Modality-Specific Normalization. Normalization parameters are fit on the training split only. For RNA-seq we apply $\mathbf{x}'_{mRNA} = \text{StandardScale}(\ln(\mathbf{x}_{mRNA} + 1))$; for CNV and methylation, $\mathbf{x}' = \text{StandardScale}(\mathbf{x})$. Median imputation, where required, is likewise fit on the training fold.

2. Architectural Formulation

2.1. Non-Linear Modality Lifting

Genomic and epigenomic variations do not scale linearly; biological mechanisms operate on physiological thresholds. For each patient and selected gene $i \in \{1, \dots, N\}$, scalar values are projected into a d -dimensional space ($d = 128$) by per-modality Feature Tokenizers (small MLPs), with learnable modality embeddings $\mathbf{E}_m, \mathbf{E}_c, \mathbf{E}_t \in \mathbb{R}^d$ encoding biological origin:

$$\mathbf{z}_{m,i} = \text{MLP}_{mRNA}(x_{m,i}) + \mathbf{E}_m \quad (3)$$

$$\mathbf{z}_{c,i} = \text{MLP}_{CNV}(x_{c,i}) + \mathbf{E}_c \quad (4)$$

$$\mathbf{z}_{t,i} = \text{MLP}_{methy}(x_{t,i}) + \mathbf{E}_t \quad (5)$$

Each MLP is Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU $\rightarrow$ Linear.

2.2. Intra-Gene Fusion and Symmetrical Masking

To model local epigenetic–transcriptomic interactions (e.g. methylation silencing of mRNA), we apply local cross-attention inside each gene. The input sequence for gene i prepends a learnable token $\mathbf{z}_{cls}$:

$$\mathbf{X}_i = \begin{bmatrix} \mathbf{z}_{cls} \\ \mathbf{z}_{m,i} \\ \mathbf{z}_{c,i} \\ \mathbf{z}_{t,i} \end{bmatrix} \in \mathbb{R}^{4 \times d}. \quad (6)$$

Unimodal Dropout. Masking a single modality enforces cross-modal redundancy. With probability p_{drop} at most one channel is masked:

$$\tilde{\mathbf{z}}_{k,i} = \begin{cases} 0 & \text{now then } \mathbf{z}_{mask}^{(k)}, \\ \mathbf{z}_{k,i}, & \text{otherwise} \end{cases} \quad \text{prob. } p_{\text{drop}} \quad \forall k \in \{m, c, t\}, \quad (7)$$

where $\mathbf{z}_{mask}^{(k)} \in \mathbb{R}^d$ is a learnable per-modality mask token; capping at one masked channel lets the model impute the missing modality from the remaining two. Because RNA-seq is the most information-dense modality and tends to dominate, a higher dropout rate is applied to it specifically to prevent the model from collapsing onto transcriptomics. **[TODO: Change 0 to $\mathbf{z}_{mask}^{(k)}$ when the masking token will be implemented and tested]**

Intra-Gene Attention. Multi-head self-attention with a Pre-LayerNorm formulation,

$$\mathbf{X}'_i = \mathbf{X}_i + \text{MHA}(\text{LN}(\mathbf{X}_i)), \quad (8)$$

produces the unified gene representation $\mathbf{h}_i = \mathbf{X}''_{i,0} \in \mathbb{R}^d$ from the updated classification token.

2.3. Graph Positional Encoding

Beyond the attention bias below, each gene token is endowed with a graph-structural positional encoding computed from the induced PPI subgraph of the selected genes. We use Random-Walk Positional Encodings (RWPE) [5]: with $\mathbf{A}$ the (binary, symmetric) adjacency and $\mathbf{P} = \mathbf{D}^{-1}\mathbf{A}$ the random-walk transition matrix, the k -step return probabilities

$$\mathbf{p}_i = [\mathbf{P}^1]_{ii}, [\mathbf{P}^2]_{ii}, \dots, [\mathbf{P}^{d_{pe}}]_{ii} \in \mathbb{R}^{d_{pe}} \quad (9)$$

encode each gene’s local topological role. The encoding is exposed as a modular component (RWPE or Laplacian eigenvectors) so the positional scheme can itself be tuned.

2.4. Global Transformer and Biased Attention

To respect canonical pathways, we bias the global attention matrix with the Shortest Path Distance (SPD) of the PPI network. Each head h maps integer distances to a learnable bias, bounded by $\tanh$ to preserve gradient stability:

$$\mathbf{B}_{\text{spatial}}^{(h)} = \tanh\left(\text{Embed}^{(h)}(\mathbf{D}_{\text{SPD}})\right). \quad (10)$$

A [TUMOR_CLS] token aggregates information across all genes, $\mathbf{H}^{(0)} = [\mathbf{t}_{cls}, \mathbf{h}_1, \dots, \mathbf{h}_N]$. In the current model the structural bias is injected through separate row-wise softmax terms (a dual-softmax formulation),

$$\text{Attn}^{(h)} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} + \text{softmax}\left(\mathbf{B}_{\text{spatial}}^{(h)}\right)\mathbf{V}, \quad (11)$$

following the row-wise scaling of Graphormer-Boosted [4]. A canonical Graphormer formulation that places the bias *inside* a single softmax is a planned refinement (Sec. 6).

2.5. Classification and Dampened Focal Loss

The $K = 8$ pathway tokens of the final layer are aggregated into a systemic tumor state $\mathbf{u}_{\text{tumor}} = \text{MLP}_{\text{agg}}([\mathbf{t}_{p_1}^{(L)}, \dots, \mathbf{t}_{p_K}^{(L)}]) \in \mathbb{R}^d$, and subtype logits are produced by a bottleneck head ($d \rightarrow d/2$):

$$\hat{\mathbf{y}} = \mathbf{W}_2 \text{GELU}(\text{BatchNorm}(\mathbf{W}_1 \mathbf{u}_{\text{tumor}} + \mathbf{b}_1)). \quad (12)$$

Optimization uses a Multi-Class Focal Loss [7] with $\gamma = 2.0$, with square-root-dampened inverse class frequencies $\alpha_c = \sqrt{1/N_c}$ to counter the extreme Normal-like imbalance without inducing degenerate “safe-guess” behavior.

3. Native Interpretability

The architecture supports interpretation without external attribution methods:

- **Global driver extraction:** the final-layer attention from the pathway tokens identifies the genes most heavily attended.
- **Deep drill-down:** for each driver gene, its 4×4 intra-gene attention matrix isolates which omics modality drove the anomalous token state.

4. Evaluation Methodology

Because classical early-fusion models are strong in the $N \ll p$ regime, any claim of deep-model benefit must survive a rigorous, identical protocol applied to all methods. We therefore evaluate every model — baselines and MOGFormer — through a single shared harness.

Cross-validation. We use repeated stratified k -fold cross-validation at the patient level: 5 folds repeated 5 times for 25 estimates, stratified on PAM50 label. Five folds (rather than ten) keep ~ 7 Normal-like patients per held-out fold. Fold indices are persisted so every model is scored on identical partitions, enabling *paired* statistical tests.

Leakage controls. Within each outer fold, HVG selection, imputation, normalization, *and* the induced PPI subgraph (hence the RWPE and SPD matrices) are recomputed on the training patients only; no held-out patient influences any fitted statistic or the graph. Hyperparameter tuning and early stopping use an inner stratified 20% split of the training fold; the outer fold is never seen during model selection.

Statistics. We report macro-F1 as the primary metric (mean $\pm$ 95% CI over the 25 estimates), together with per-class F1, balanced accuracy, Matthews correlation, and per-class ROC/PR-AUC. Because cross-validation folds overlap, we additionally report the Nadeau–Bengio corrected-resampled variance [11] and, for model-versus-model claims, the paired Wilcoxon signed-rank test on the 25 paired per-fold scores, summarized through critical-difference diagrams [12].

Baseline suite. All baselines run through the identical pipeline and folds: a stratified *Dummy* floor; a PAM50 nearest-centroid classifier on RNA (the label-source reference [1]); penalized multinomial logistic regression; a linear SVM; random forests; gradient-boosted trees (XGBoost); and TabPFN-v2 [9] on the top- K genes. Each classical model is evaluated under both early fusion (concatenated standardized features) and late fusion (per-modality models with averaged class probabilities), and MOGONET is retained as a direct GNN comparator [8].

5. Results

Classical early-fusion is a demanding baseline. A well-tuned XGBoost reaches a macro-F1 of 0.852, well above the PAM50 nearest-centroid reference (0.738); since PAM50 labels are themselves RNA-derived, the centroid classifier is the most conceptually important reference point (Table 1). This sets a stringent target for any deep model.

Multi-omics versus single-omics. Across the modality-contribution grid, RNA-only performance (~ 0.86 macro-F1) is statistically indistinguishable from the full three-modality configuration, with CNV and methylation adding little *accuracy* over

Table 1: PAM50 subtyping under the shared 25-estimate protocol (early fusion, all three modalities). Baseline figures are current Phase-0 results; MOGFormer is under evaluation through the identical harness.

Model	Macro-F1
Dummy (stratified)	[TODO: floor]
PAM50 nearest-centroid (RNA)	0.738
Gradient-boosted trees (XGBoost)	0.852
MOGFormer (ours)	<i>under evaluation</i>

Figure 2 placeholder — MOGFormer macro-F1 ($\pm 95\%$ CI) against reference lines (XGBoost 0.852, PAM50-centroid 0.738), plus the per-class F1 panel emphasizing Normal-like.

Figure 2: MOGFormer versus reference models under the shared protocol, with per-class breakdown. **[TODO: Insert C00/F01 once the 25-estimate MOGFormer run completes.]**

RNA alone (Fig. 2). This motivates MOGFormer’s design emphasis on *interpretable* cross-modal routing rather than accuracy gains from fusion per se, and frames cross-modal contribution as a question best answered by attention attribution rather than by held-out F1.

Which classes drive the error. Per-class F1 for the best classical model is high for Basal (0.98), Luminal A (0.94), HER2 (0.89), and Luminal B (0.88), but collapses for Normal-like (0.51). The dominant confusions are Luminal A $\leftrightarrow$ Luminal B and Normal-like $\rightarrow$ Luminal A. Normal-like ($n = 35$) is therefore the universal bottleneck, and we report per-class F1 at every step to prevent the macro average from masking this collapse.

Feature-count sensitivity. Performance is stable for $N \in [100, 500]$ selected genes, defining the operating regime; we fix $N = 250$ to balance this plateau against the $O(N^2)$ cost of global attention.

6. Ongoing Refinements and Future Work

6.1. Targeted Architectural Refinements

Guided by the gap to the classical baseline, four single-variable refinements are under controlled evaluation, each applied and measured in isolation through the shared harness: (i) a per-gene identity embedding $\mathbf{G}_i$ added to every modality token, so the value path can distinguish genes of equal scaled magnitude; (ii) a canonical Graphormer attention that places the structural bias inside a single softmax rather than the dual-softmax of Sec. 2.4; (iii) periodic (PLR) numerical embeddings [6] replacing the per-modality MLP tokenizers; and (iv) the learnable per-modality mask token for unimodal dropout. **[TODO: Report deltas once each single-change run completes; these are hypotheses, not yet results.]**

6.2. Self-Supervised Pan-Cancer Pre-Training

To mitigate the $N \ll p$ constraint, we plan a self-supervised pre-training stage on unlabeled pan-cancer TCGA cohorts, building an oncogenic foundation representation before downstream adaptation. A random 15% subset $\mathcal{M}$ of genes has its three modalities replaced by a learnable mask token, $\mathbf{h}_i^{(0)} = \mathbf{z}_{\text{mask}} \forall i \in \mathcal{M}$; the masked sequence is propagated through the structurally biased layers and a regression head reconstructs the original normalized values, optimized by a masked MSE,

$$\mathcal{L}_{MGM} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \sum_{k \in \{m, c, t\}} (x_{k,i} - \hat{x}_{k,i})^2. \quad (13)$$

After convergence, the reconstruction head is discarded, the tokenizers, intra-gene mini-transformers, and lower global layers are frozen, and the pathway tokens plus top classification layers are fine-tuned with the dampened focal loss — reducing the sample complexity required on rare subtypes. **[TODO: This section describes planned work; mark clearly as such or omit for a results-focused submission.]**

Methods

Graph Construction and Spatial Bias Embedding

The biological prior is the Human PPI network from STRING [10]. Let $G = (V, E)$ with V the per-fold selected genes (N) and E the interactions with combined score ≥ 400 , mapped from STRING protein identifiers to HGNC symbols via the alias table. The graph is induced on the selected genes and rebuilt per fold. The Shortest Path Distance matrix $\mathbf{D}_{SPD} \in \mathbb{R}^{N \times N}$ is computed with a cutoff D_{max} ; unreachable pairs are assigned $D_{\text{max}} + 1$. Each integer distance maps to a learnable per-head scalar bias bounded by tanh.

Model Optimization and Hyperparameters

MOGFormer is trained with AdamW ($\eta = 10^{-4}$, weight decay 10^{-4}), token dimension $d = 128$, $K = 8$ latent pathway tokens, and the Multi-Class Focal Loss ($\gamma = 2.0$, square-root-dampened class weights). Early stopping monitors inner-validation macro-F1. **[TODO: Align the following to the code you are running: the current configuration uses $L = 1$ global transformer layer and $D_{\text{max}} = 10$]**

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